

KaraokAI: Exploring the Potential of AI-Generated Karaoke for Academic Conferences and Research Experiences

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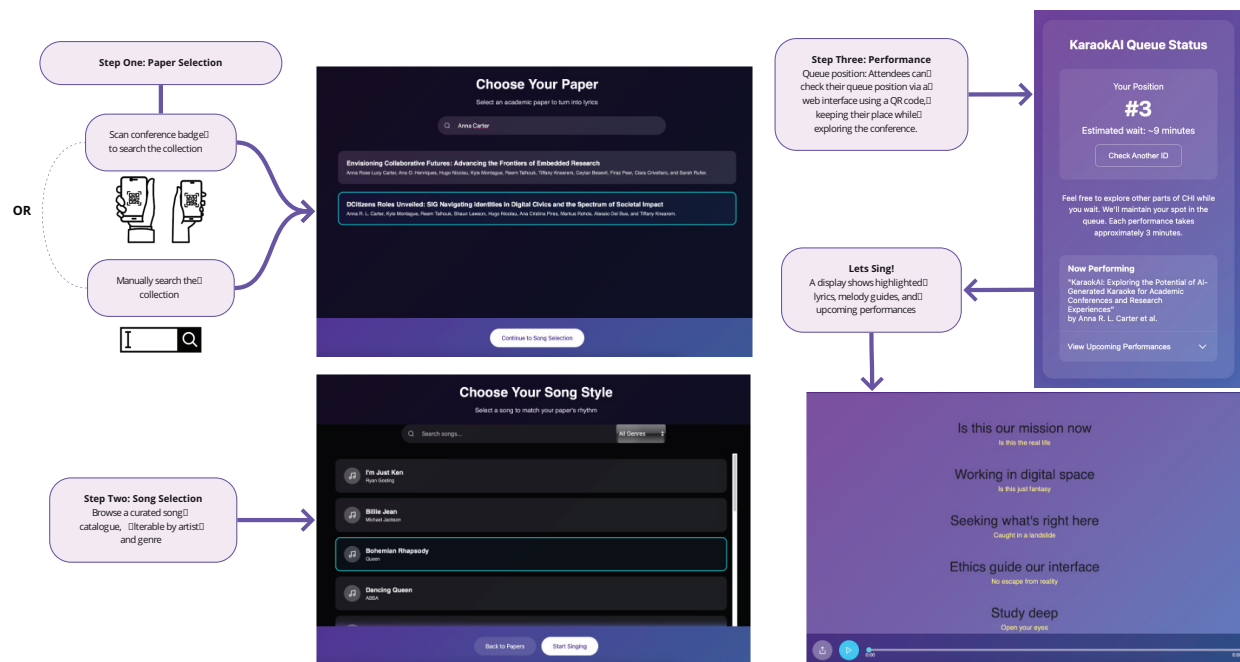


Figure 1: Diagram showing the user interaction experience of KaraokAI

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Abstract

Academic conferences face increasing challenges as Generative AI accelerates academic writing, dramatically increasing submission volumes and straining reviewer capacity. Simultaneously, shifts in knowledge consumption, driven by AI tools and multimedia platforms, highlight the need for more engaging dissemination formats. Our interactive demonstration, KaraokAI, is a karaoke-based presentation platform which transforms academic papers into lyrics

performed alongside popular songs. Drawing on the participatory nature of karaoke, this approach enhances engagement, recall, and emotional connection, making research dissemination more accessible and memorable. With CHI 2025 being held in Japan, the birthplace of karaoke, this demonstration explores how cultural and playful methods can reimagine academic presentations, inviting collaborative and inclusive participation. By blending scholarship with entertainment, KaraokAI aims to challenge traditional dissemination norms and inspire innovative, audience-centred approaches to sharing research across diverse contexts. So, grab the mic, hit the stage and let your research take the spotlight.

CCS Concepts

• **Music-based Learning**; • **Alternative Education**; • **Knowledge Exchange**; • **Engagement**; • **Interactivity**;

Keywords

HCI, Music, LLM

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1 Introduction & Background

Academic conferences have been a cornerstone in the community since the late 18th century [5] aiming to (a) bring people together to disseminate knowledge, (b) foster collaborations across disciplines, and (c) explore a variety of perspectives. However, the landscape of academic dissemination is changing. The rise of Generative AI technologies has increased the ease and speed of academic writing [18], creating unprecedented challenges in managing the sheer volume of submissions for major conferences. For example, CHI has experienced a 94% increase in paper submissions since 2022 [27] – the same year ChatGPT was released [19]. Prior to this, submission rates had increased by roughly 10% a year, suggesting that the emergence of tools like ChatGPT may have significantly accelerated the academic writing process and contributing to this dramatic surge. This year alone, there were 5,014 papers were submitted to CHI, each requiring two reviewers and two associate chairs – totalling four reviewers per paper, and 1,969 papers were given the opportunity to revise and resubmit [27]. This will have resulted in a minimum of 27,932 reviews for CHI 2025. Such significant growth at these major conferences not only adds to the already substantial burden on the overextended volunteer reviewer pool but also raises critical questions about how we can sustainably share and engage with research.

At the same time, we are witnessing fundamental shifts in the way people consume and engage with knowledge. The emergence of AI-powered tools for literature reviews and writing [3, 6], research synthesis platforms such as NotebookLM [14], and the growing preference for multimedia content suggests an opportunity to expand beyond traditional paper presentations to embrace diverse

learning and engagement styles. Educational technologies have long leveraged entertainment formats to enhance learning, from musical educational games to interactive learning apps [15, 20]. Social media apps have demonstrated the potential of educational content delivered through creative formats, such as dancing the news [24] or educating on climate change [4], highlighting the shift towards a more dynamic and engaging way to sharing knowledge [9, 17]. This shift in how we consume knowledge opens up exciting new possibilities. Drawing inspiration from object-based media principles, where content adapts to the device, user, and context [13], we can reimagine research dissemination to be just as flexible and contextually aware.

While conferences like CHI have explored alternate presentation methods, such as technology demonstrations, discussion-based sessions, and alt.chi which encourages unconventional formats and provocative forms of dissemination, the core approaches to research dissemination have remained almost the same [11]. However, effectively engaging diverse audiences and stakeholders often requires tailoring how research is presented. At its core, research dissemination is a human-to-human process of sharing knowledge and insights. While technology can help transform and adapt content into different formats, the essence of research communication lies in how humans interpret, perform, and connect with the material. By embracing this human-in-the-loop approach, we can leverage technology to enhance rather than replace the inherently human aspects of knowledge sharing, creating opportunities for more dynamic and interactive engagement with knowledge.

Research demonstrates that multimedia presentations can significantly increase listener engagement [25], and that music can serve as a powerful memory technique where information is associated with familiar melodies [8, 16]. Just as songs and rhymes help children learn the alphabet or sequences [21], presenting research as lyrics to popular tunes can make findings more memorable [7], while deepening understanding through emotional and embodied responses [7, 26].

This demonstration introduces KaraokAI, a karaoke-based platform that transforms academic papers into lyrics performed alongside well-known songs. Inspired by the collective, inclusive and participatory nature of karaoke [2, 10, 28], KaraokAI aims to reimagine research dissemination. With CHI 2025 taking place in Japan, the birthplace of karaoke and home to more than 8,000 karaoke bars [1], the context is particularly fitting for this exploration.

By transforming research papers into “research-songs”, participants can present their work in a playful and memorable format that fosters collaboration, by performing themselves, inviting others to join or observing a research song presentation. This approach invites active engagement and encourages reflective discussions on the evolution of research sharing. These performances can extend beyond the conference, reaching classrooms, lab meetings, social events and public audiences. Through this demonstration, we aim to challenge traditional academic presentations and explore how integrating cultural, interactive formats like karaoke can make research more accessible, engaging and memorable.

2 Interactive Experience

KaraokAI is an open source web-based karaoke system designed to transform academic papers into karaoke songs, using familiar melodies to engage and entertain conference attendees. Building on familiar karaoke interfaces while integrating with the research environment, the system provides an interactive performative experience, enabling attendees to explore research in a playful and memorable way through the following steps (with accompanying interface representations shown in Figure 1):

Step One: Paper and Song Selection When approaching the *KaraokAI* station, attendees can either scan their conference badge to instantly access papers they are associated with, or explore the full collection of available papers through a searchable interface. After selecting a paper, attendees can browse through our song catalogue, which can be filtered by artist and genre to find the perfect match for their chosen paper. Songs were selected based on (a) their global recognisability, informed by top karaoke song lists [12, 22, 23] and personal knowledge, (b) their availability as karaoke tracks and, (c) the availability of timestamped lyrics.

Step Two: Queue Management Once selections are confirmed, attendees receive a unique performance ID indicating their position in the performance queue. While attendees wait their turn, the system works in the background to generate unique lyrics inspired by and based on the selected paper, and matched to the chosen song’s melody and rhythm. This parallel processing makes efficient use of queue time, ensuring performances can flow smoothly. Through a dedicated web interface, attendees can check their queue position from their own web-enabled devices. The queuing system gives the attendees the flexibility to stay and enjoy the performances or to explore other parts of the conference while maintaining their place in line, ensuring participants can make the most of their conference experience without missing their performance slot.

Step Three: Performance The main demonstration area features a display showing the current performance’s lyrics in traditional karaoke style (also visible on the participant’s mobile device), with each line highlighted in time with the music. Beneath each line, the original song lyrics are displayed as a melody guide, helping singers stay on rhythm even when performing the newly generated lyrics for the first time. The display also indicates the upcoming performance, helping attendees track when their turn is approaching. The *KaraokAI* station’s operators can manage the queue through a control interface, accommodating schedule changes or late arrivals as needed.

To accommodate different comfort levels with public performances, our demo station will provide options for both headphones and speaker output. This allows attendees to choose between a more private session or a full performance that might draw in other conference attendees, creating spontaneous moments of shared engagement. To ensure the experience extends beyond the conference, attendees retain access to their generated lyrics through their unique performance

ID, allowing them to revisit and perform their research-backed karaoke song at any time. Through *KaraokAI*, we aim to showcase a new way of presenting in a playful and engaging way, reimagining how insights can be shared and experienced in a universally enjoyable format.

3 System Architecture

KaraokAI follows a modular pipeline approach for transforming academic content into performative experiences as shown in Figure 2. While this implementation focuses on karaoke as the output format, the pipeline and object-based approach could be adapted for various forms of research dissemination in the future (e.g., Slam Poetry, Rap Battle and Theatrical Monologues).

The system’s back-end consists of three main components: *song pre-processing*, *paper analysis*, and *lyrics generation phases*. In the preprocessing stage, we analyse the curated song collection to create structural “blueprints” for each track. This involves identifying syllable counts per line, rhyming patterns, and repeated sections (e.g., chorus sections). We create timestamp mappings to ensure synchronisation between lyrics and music. These blueprints are stored to be used as structural templates for the lyrics generation.

Each paper undergoes automated analysis using a Large Language Model (LLM) to extract key research motivations, methodologies, findings and contributions. This structured analysis is stored for subsequent use, enabling efficient retrieval and faster generation when the same paper is selected in future interactions.

The final lyrics generation phase occurs when a song and paper pair are selected and queued for performance. The system combines the paper’s analysed content with the song’s structural blueprint as input to an LLM, using prompts designed to specify both lyrical constraints and research communication requirements. This ensures the generated lyrics maintain the original song’s rhythm and rhyme patterns while effectively communicating the research content. The resulting lyrics are then synchronised with the karaoke backing track using the pre-generated timestamps.

The system’s web-based front-end allows paper and song selection, monitoring and managing of the performance queue, and playback of the music track with a synchronised display of the generated lyrics.

Through this pipeline, *KaraokAI* creates an experience that aims to ensure responsive performance during the interactivity demonstration, allowing attendees to quickly generate and perform their research-based karaoke songs.

4 Relevance to CHI

Karaoke is more than just entertainment; it’s a social experience that brings people together across cultures, fostering a sense of shared purpose. Similarly, by blending scholarly dissemination with an interactive and performative method, we offer attendees an opportunity to engage with research in a new way, turning the presentation into a shared experience. This approach reflects practical benefits in today’s research landscape. This alignment with Ikigai, CHI2025’s theme, underscores how innovative, culturally-infused formats not only foster collaboration and connection but also provide a path toward more sustainable and efficient research dissemination.

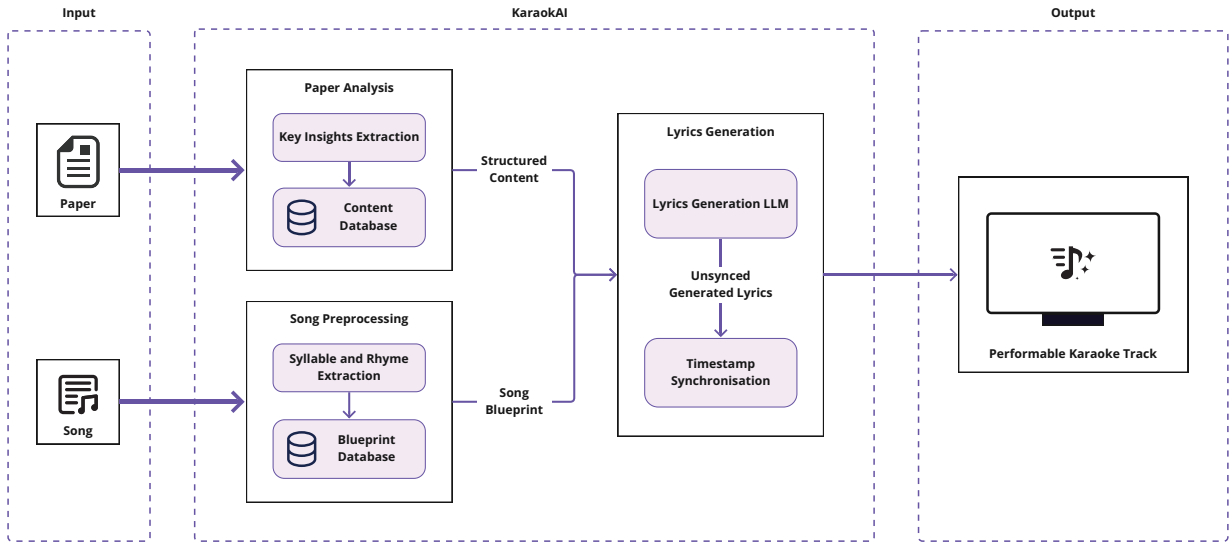


Figure 2: KaraokAI system diagram showing the inputs, processing steps, and output.

5 Reflections and Future Directions

Our current implementation of KaraokAI represents an initial exploration into creative research dissemination, while suggesting several directions for future development. Our decision to focus on English-language songs and papers, was influenced both by practical constraints and CHI’s primary publication language, highlights broader questions about accessibility and cultural representation in academic knowledge disseminations. Similarly, While our song selection draws from widely recognised tracks, we acknowledge the limitations in our catalogue, which primarily reflects Western popular music. Despite these constraints, the demonstration serves as a proof of concept, showcasing potential for future expansions into other languages and cultural contexts.

The transformation pipeline we developed for KaraokAI demonstrates one possible approach to reimagining research dissemination. The same principles of analysing and restructuring academic content could be applied to create various alternative formats, from comic strips to short-form videos. Each format would require its own structural blueprints and transformation rules, much like KaraokAI’s syllabic patterns for musical adaptation, but would follow similar principles of content analysis and creative transformation.

Looking beyond format transformation, we can see broader possibilities for personalised research experiences. Future iterations could adapt the research content analysis based on audience expertise - whether researchers from different disciplines, members of the public, or school children as part of research outreach. Furthermore, by creating engaging, shareable versions of research content, systems like KaraokAI could help bridge the gap between academic and public domains, making research more accessible on social media and other public platforms.

6 Conclusion

We present KaraokAI, an open-source karaoke system that leverages generative AI to transform academic papers into performable songs, offering a playful and engaging approach to research dissemination. Through this interactive demonstration, attendees can experience how traditional academic content can be reimagined into memorable, shareable formats. Beyond just entertainment, this system serves as a proof-of-concept for how we might adapt to increasing conference submissions and changing patterns of knowledge consumption. By combining the familiar social experience of karaoke with academic content, we hope to spark discussions about the future of research presentations and how we can make them more accessible, engaging, and memorable for diverse audiences.

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Generative AI was used to create the following sections of this work: Anthropic’s Claude as part of the lyric generation system (as described above), including the example lyrics in Figure 1 and the accompanying video. Suno AI Music Generator for the music and vocals for the accompanying video.

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